The Spanish Node of Federated EGA

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The Federated European Genome-Phenome Archive (FEGA) (*see POSTER #28*) is the primary European resource for discovering and accessing sensitive human -omics- data consented for secondary use. The network comprises national repositories that ensure data remains in local jurisdictions while being discoverable through the main EGA portal and accessed across international borders in a federated manner.

The Spanish National Bioinformatics Institute (INB), as the ELIXIR node in Spain, and in collaboration with the EGA team at the Centre for Genomic Regulation (CRG), has established the Spanish Federated EGA node (ES-FEGA), primarily hosted at BSC facilities.

Data Submission

Legal Agreements

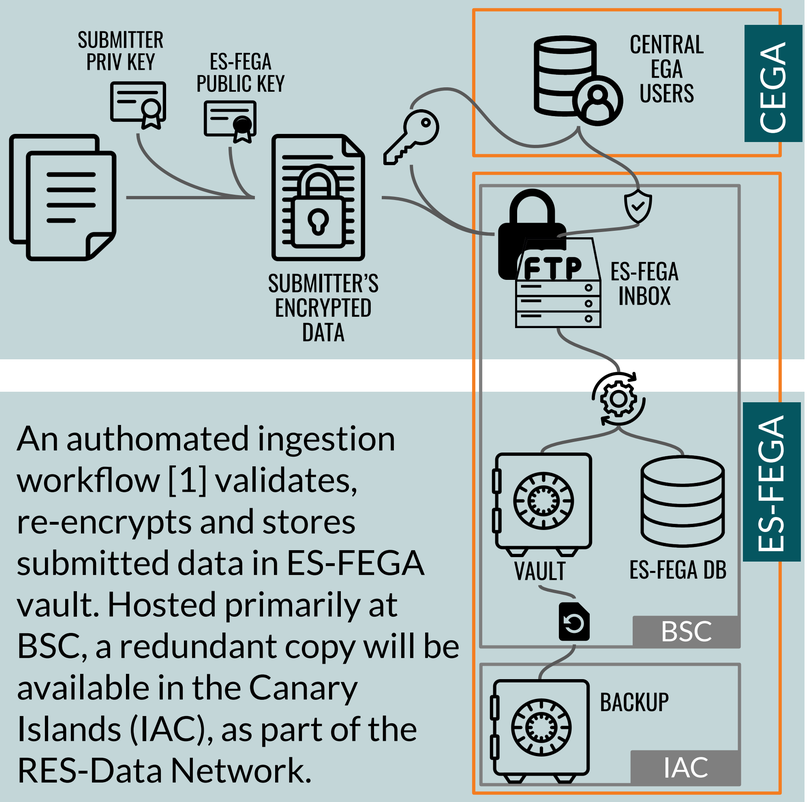
Submitter confirms its legal authority for depositing data

DPA and other legal documentation is signed being complying with GDPR and local spanish regulation

Data Encryption & upload

Submitter encrypts and loads data to FEGA inbox

Submitter encrypts the data using its GA4GH Cryp4GH private key and load it via SFTP to ES-FEGA inbox using Central EGA credentials.



Sensitive data archived in Spain

Data is primarily hosted at BSC

An authomated ingestion workflow [1] validates, re-encrypts and stores submitted data in ES-FEGA vault. Hosted primarily at BSC, a redundant copy will be available in the Canary Islands (IAC), as part of the RES-Data Network.

Metadata submission

Public metadata is submitted to central EGA

Publication to Central EGA

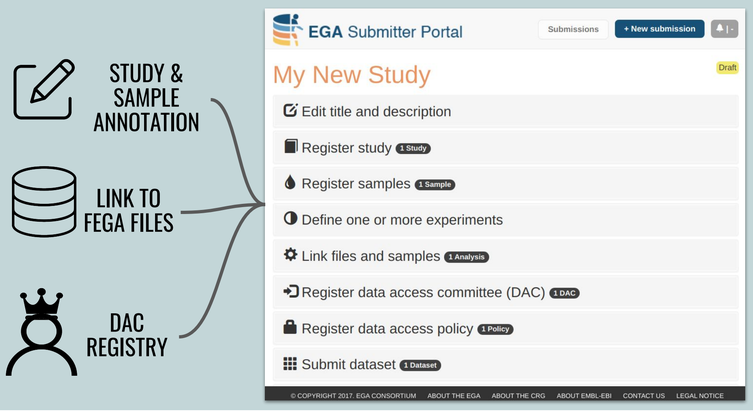
New dataset is publicly available

Data Retrieval

Data discovery

Requester browses central EGA catalogue

Submitter fills in the details on the study at the central EGA Submitter Portal



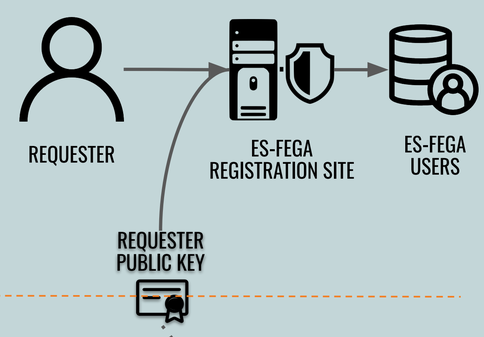
Requester browses central EGA catalogue

Data access management

DAC agrees on sharing

Requester and DAC representative agree on a DAA for sharing the dataset/s of interest.

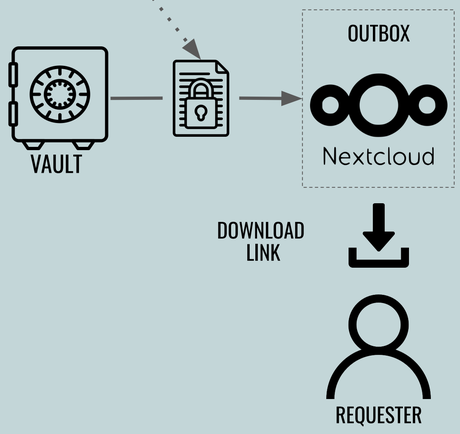
Requester registers in ES-FEGA, providing its cryp4gh public key, which later will be used to re-encrypt requested data.



Data delivery

Requester downloads dataset from ES-FEGA outbox

ES-FEGA Helpdesk provides the requester with an ES-FEGA outbox download link [2] granting access to the requested datasets, re-encrypted with the requester key.



ACRONYMS

BSC: Barcelona Supercomputing Center
CRG: Centre for Genomic Regulation
DAA: Data Access Agreement
DAC: Data Access Committee
DPA: Data Processing Agreement
EGA: European Genome-phenome Archive
IAC: Instituto de Astrofísica de Canarias
INB: Spanish National Bioinformatics Institute
GDPR: General Data Protection Regulation
RES: Spanish Supercomputing Network

USEFUL REFERENCES

1. **Local EGA repository:** https://github.com/EGA-archive/LocalEGA
2. **ES-FEGA outbox data delivery:** https://gitlab.bsc.es/inb/elixir/ega/fega-nextcloud-transport

Federated European Genome-phenome Archive SPAIN

Contact

ES-FEGA Helpdesk
es-fega.helpdesk@bsc.es

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